Notes on Schur Decomposition

Handwritten Notes Conversion

Introduction

When I studied Gilbert Strang's *"Linear Algebra and Its Applications"*, his explanation of Schur Form was not very clear. This led me to find out more about Schur decomposition to the point where I could implement it.

1 Theorem Statement

Theorem 1 (Schur Decomposition)**.** *Let $A \in \mathbb{C}^{n\times n}$. There exists a unitary matrix $U \in \mathbb{C}^{n\times n}$ and an upper triangular matrix $T \in \mathbb{C}^{n\times n}$ such that:*

$$U^{-1}AU = T \quad \iff \quad A = UTU^{\dagger}$$

2 Proof and Method by Induction

2.1 Base Case

For $A \in \mathbb{C}^{1\times 1}$, $A = [a_{11}]$.

- Choose $U_1 = [1]$, its inverse is $U_1^{-1} = [1]$.
- Compute $U_1^{-1}AU_1 = [1][a_{11}][1] = [a_{11}] = T$.
- A 1×1 matrix is trivially upper triangular.

2.2 Inductive Hypothesis

If any $(n-1)\times(n-1)$ matrix A_2 can be decomposed using a unitary matrix W_2 such that:

$$W_2^{-1}A_2W_2 = T_{n-1}$$

where T_{n-1} is an upper triangular matrix.

2.3 Inductive Step

Let A be an arbitrary $n \times n$ matrix. We must reduce it to size $(n-1)\times(n-1)$ and link it to our hypothesis.

1. We use the Power Method (or general eigenvalue analysis) to find a dominant eigenvalue λ_1 and its corresponding non-zero eigenvector $\vec{u}_1$ such that $A\vec{u}_1 = \lambda_1\vec{u}_1$.
2. We use the Gram-Schmidt process to expand our eigenvector $\vec{u}_1$ to a set of orthonormal vectors $\{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_n\}$.
3. Construct U_1: We place these vectors as the columns of U_1:

$$U_1 = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \ldots & \vec{u}_n \end{bmatrix}$$

We want to prove that $U_1^{-1}AU_1$ has zeros everywhere in the first column below λ_1.

$$AU_1 = \begin{bmatrix} A\vec{u}_1 & A\vec{u}_2 & \dots & A\vec{u}_n \end{bmatrix}$$

When a matrix multiplies another matrix, it acts on each column individually. For the first column, we get:

$$A\vec{u}_1 = \lambda_1 \vec{u}_1$$

For the other columns ($A\vec{u}_2$ through $A\vec{u}_n$), the outputs are vectors in space. Because $\{\vec{u}_1, \dots, \vec{u}_n\}$ forms a basis, any vector can be written as a linear combination of them:

$$A\vec{u}_j = x_{1j}\vec{u}_1 + x_{2j}\vec{u}_2 + \cdots + x_{nj}\vec{u}_n$$

Notice that:

$$U_1 \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \dots & \vec{u}_n \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = c_1\vec{u}_1 + c_2\vec{u}_2 + \cdots + c_n\vec{u}_n$$

This shows that multiplying U_1 by a column vector of constants is the same as making a linear combination of the columns of U_1.

- For the 1st column: $\lambda_1\vec{u}_1 + 0\vec{u}_2 + \cdots + 0\vec{u}_n$. The column vector of numbers we need to multiply U_1 by is $\begin{bmatrix} \lambda_1 & 0 & \dots & 0 \end{bmatrix}^T$.
- For the rest of the columns, $(A\vec{u}_j)$ corresponds to $\begin{bmatrix} x_{1j} & x_{2j} & \dots & x_{nj} \end{bmatrix}^T$.

Putting them side by side forms:

$$AU_1 = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \dots & \vec{u}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & x_{12} & \dots & x_{1n} \\ 0 & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & x_{n2} & \dots & x_{nn} \end{bmatrix}$$

So this is:

$$AU_1 = U_1 \begin{bmatrix} \lambda_1 & c_1^T \\ \mathbf{0} & A_2 \end{bmatrix}$$

Multiplying by U_1^{-1}:

$$U_1^{-1}AU_1 = \begin{bmatrix} \lambda_1 & c_1^T \\ \mathbf{0} & A_2 \end{bmatrix}$$

The inner submatrix A_2 has dimension $(n-1) \times (n-1)$. By our inductive hypothesis, we can recursively decompose A_2. There exists an $(n-1) \times (n-1)$ unitary matrix W_2 such that $W_2^{-1}A_2W_2 = T_{n-1}$ is upper triangular.

Define:

$$U_2 = \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & W_2 \end{bmatrix}, \quad U_2^{-1} = \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & W_2^{-1} \end{bmatrix}$$

We substitute into $U_2^{-1}(U_1^{-1}AU_1)U_2$:

$$\begin{aligned} U_2^{-1}(U_1^{-1}AU_1)U_2 &= \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & W_2^{-1} \end{bmatrix} \begin{bmatrix} \lambda_1 & c_1^T \\ \mathbf{0} & A_2 \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & W_2 \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 & c_1^T W_2 \\ \mathbf{0} & W_2^{-1}A_2W_2 \end{bmatrix} \end{aligned}$$

Since $W_2^{-1}A_2W_2 = T_{n-1}$:

$$U_2^{-1}(U_1^{-1}AU_1)U_2 = \begin{bmatrix} \lambda_1 & c_1^T W_2 \\ \mathbf{0} & T_{n-1} \end{bmatrix} = \text{upper triangular matrix}$$

The final upper triangular matrix T is:

$$T = \begin{bmatrix} \lambda_1 & c_1^T W_2 \\ \mathbf{0} & T_{n-1} \end{bmatrix}$$

Since T_{n-1} is upper triangular, this keeps the full matrix T upper triangular.

The full unitary transformation matrix is:

$$U = U_1U_2U_3 \dots U_n$$

This is our final unitary matrix U.

Since we have proved that an $n \times n$ matrix can be decomposed if an $(n-1) \times (n-1)$ matrix can, and our base case works, we have proven Schur Decomposition by induction and derived a method to compute it.

3 Properties of the Unitary Product and Eigenvalues

To prove that the product of unitary matrices is unitary, note that for U_1, U_2:

$$(U_1U_2)^H = U_2^H U_1^H = U_2^{-1}U_1^{-1} = (U_1U_2)^{-1}$$

because $U_2^{-1}U_1^{-1}U_1U_2 = I$.

3.1 Eigenvalues on the Diagonal

The eigenvalues of A are on T's diagonal. This is because A's eigenvalues and T's eigenvalues are the same, and T's eigenvalues lie on its diagonal.

Evaluating $\det(T - \lambda I)$:

$$\det(T - \lambda I) = \det \begin{bmatrix} \lambda_1 - \lambda & & * \\ & \lambda_2 - \lambda & \\ 0 & & \ddots \end{bmatrix} = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \dots (\lambda_n - \lambda)$$

Since the terms below $(\lambda_1 - \lambda)$ are 0, the determinant simplifies directly to the product of diagonal entries.

To prove that A's eigenvalues are T's eigenvalues:

$$\begin{aligned} \det(A - \lambda I) &= \det(UTU^{-1} - \lambda UU^{-1}) \\ &= \det(U)\det(T - \lambda I)\det(U^{-1}) \\ &= \det(UU^{-1})\det(T - \lambda I) \\ &= \det(T - \lambda I) \end{aligned}$$

Because A's eigenvalues are equivalent to T's eigenvalues, and T's eigenvalues lie on its diagonal, A's eigenvalues can be read directly off T's diagonal.